

Ch.3. Welding

Focus: Introduction to Welding, Arc Welding, Oxy - Acetylene Welding, Brazing, and Soldering

1. Introduction to Welding

Welding is a fundamental skill in metal fabrication, joining materials through heat, pressure, or both, often with a filler material. It is vital in industries like construction, automotive, aerospace, and manufacturing. This guide covers essential techniques, equipment, and safety practices for beginners, focusing on arc welding, oxy-acetylene welding, brazing, and soldering.

Key Categories of Joining (Welding) Processes:

1. Arc Welding (SMAW): Uses an electric arc between a consumable electrode and the work piece. Common for construction and repairs.
2. Oxy-Acetylene Welding: Combines acetylene gas and oxygen to create a flame (3,200°C). Ideal for cutting and thin metals.
3. Brazing: Joins metals using filler (e.g., brass) melted above 450°C but below work piece melting point (but base metals remain solid).
4. Soldering: Low-temperature process (<450°C) with tin-lead filler for electronics and plumbing.

2. General Safety Precautions

- PPE (Personal Protective Equipment): Helmets, gloves, flame-resistant clothing, safety glasses, and boots.
- Ventilation: Avoid inhaling fumes; use exhaust systems.
- Fire Safety: Keep extinguishers nearby; clear flammable materials.
- Gas Handling: Secure cylinders upright, check for leaks with soapy water.
- Training: Never operate equipment without supervision.

3. Types of Welding Joints

1. Butt Joint: Edges aligned in the same plane.
2. Lap Joint: Overlapping pieces.
3. Tee Joint: Perpendicular pieces (T-shape).
4. Corner Joint: Edges meet at an angle.
5. Edge Joint: Flanged edges welded along seams.

Welding positions

3. Arc Welding

Definition: Uses an electric arc between an electrode and workpiece to generate heat (~3,500°C).

Equipment:

- Power Source: AC/DC machines (e.g., transformer, inverter).
- Electrodes: Consumable (SMAW) or non-consumable (GTAW).
- Cables, Ground Clamp, Electrode Holder.

Arc Welding Electrodes (AWS Classification)

- E6013: Rutile coating, easy arc ignition, all positions, general-purpose.
- E7018: Low-hydrogen coating, high strength, requires storage in a dry environment.
- Decoding: E = Electrode, 60 = 60,000 psi tensile strength, 1 = all positions, 3 = coating type.

Coated electrode materials and its Functions

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Processes:

1. Shielded Metal Arc Welding (SMAW): Common for steel; uses flux-coated electrodes.
2. Gas Metal Arc Welding (GMAW/MIG): Continuous wire feed with shielding gas.
3. Gas Tungsten Arc Welding (GTAW/TIG): Precision work with non-consumable tungsten electrode.

Steps (Starting Arc Welding):

1. Prepare Workpiece: Clean workpiece, clamp securely.
2. Set current (Amperage) (AC/DC): Adjust based on electrode size (e.g., 1/8" electrode ≈ 90-130A) and material.
3. Strike the arc by
 - a. Scratch Method: Drag electrode like a match.
 - b. Tap Method: Tap electrode on workpiece and lift and
4. Maintain Arc: Keep 1/8" gap, 15–20° travel angle and consistent travel speed.
5. Deposit filler metal (if needed).

Applications: Structural steel, pipelines, shipbuilding, Machinery repair.

Advantages: Deep penetration, strong joints.

Disadvantages: Spatter, UV radiation risks.

Safety:

- o Electrical shock prevention.
- o UV protection (auto-darkening helmets).

4. Oxy-Acetylene Welding (OAW)

Definition: Uses oxygen and acetylene gases to produce a flame (~3,200°C).

Equipment:

- o Cylinders: Oxygen (green) and acetylene (maroon).
- o Regulators, Hoses, Torch, Tips.

Flame Types:

1. Neutral Flame (1:1 ratio): Balanced, used for most welding.
2. Oxidizing Flame (excess oxygen): For cutting metals.
3. Carburizing Flame (excess acetylene): For brazing or welding alloys.

Process:

1. Adjust flame (neutral, carburizing, oxidizing).
2. Preheat base metal, then add filler rod.

Applications: Repair work, thin metals, automotive.

Advantages: Portable, no electricity needed.

Disadvantages: Slower, less penetration.

Safety:

- Prevent gas leaks (soapy water test).
- Use flashback arrestors.

5. Brazing

Definition: Joins metals using filler (e.g., brass/silver) melted above 450°C.

Equipment: Torch (furnace, or induction heater), flux, filler rods.

Filler rod: Brass, silver alloys.

Flux: Borax-based to prevent oxidation.

Steps:

1. Clean surfaces.
2. Apply flux to prevent oxidation.
3. Heat joint until filler flows via capillary action.

Applications: Pipe fittings, HVAC, jewelry, bicycle frames.

Advantages: Strong, works with dissimilar metals.

Disadvantages: Requires tight fit-up.

Safety: Avoid flux fumes; use ventilation.

6. Soldering

Definition: Joins metals with filler (tin-lead, lead-free) below 450°C.

Equipment: Soldering iron (torch, or gun), flux, solder wire.

Flux: Rosin or acid-based.

Filler: Tin-lead or lead-free alloys.

Steps:

1. Clean and flux surfaces.
2. Heat joint and apply solder.

Applications: Electronics, plumbing, jewelry, circuit boards.

Advantages: Low heat, reversible.

Disadvantages: Weak joints, not for load-bearing.

Safety: Avoid lead exposure; wash hands post-use.

Note: Brazing and Soldering

- **Principle:** Joins metals using a filler material “without” melting the base metals.
- **Key Difference:** Soldering uses lower temperatures and weaker joints than brazing.

7. Comparison of Processes

Process	Temp Range	Strength	Materials
Arc Welding	Very High	High	Steel, Stainless
Oxy-Acetylene	High	Moderate	Thin metals, Cast Iron
Brazing	Medium-High	Moderate	Dissimilar metals
Soldering	Low	Low	Electronics, Copper

8. Conclusion

- Arc Welding: High-strength, versatile.
- Oxy-Acetylene: Portable, flexible.
- Brazing/Soldering: Lower heat, for delicate or dissimilar joints.
- Practice: Essential for skill development; always prioritize safety.

Next Steps: Hands-on practice with supervision, focusing on technique and safety protocols.

9. Example Arc Welding Exercise: Different types of welding Joint

Steps:

1. Prepare: Clean plates with a wire brush.
2. Clamp: Secure plates in overlapping position.
3. Welding: Strike arc at edge, maintain 70° angle, run a steady bead along the joint.
4. Inspect: Check for uniformity, no cracks/porosity. Grind excess if needed.

Common Defects:

- Undercut: Groove at weld edge → Reduce amperage or speed.
- Porosity: Gas pockets → Clean workpiece better.